Study on AI data center infrastructure sustainable deployment and standardization

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***Abstract*—With the rapid advancement of Artificial Intelligence (AI) technologies, their applications are expanding across increasingly diverse domains. As critical infrastructure supporting AI operations, data centers face significant challenges posed by escalating server power densities and the growing demand for reliability and energy efficiency in different AI application scenarios. This paper systematically investigates sustainable architectural frameworks and core subsystems for Artificial Intelligence Data Centers (AIDCs), including high-efficiency power supply systems, advanced cooling architectures, and intelligent monitoring platforms. The study further introduces sustainable deployment methodologies for AIDC subsystems. Additionally, it conducts a comprehensive review of international standards governing AIDC development, as promulgated by major Standards Development Organizations (SDOs). The findings establish technical foundations and operational guidelines for the construction of sustainable, high-performance AI infrastructure aligned with global standardization initiatives.**

***Keywords*—Artificial Intelligence, data center, energy efficiency, power supply system, cooling system, monitoring system**

I. Introduction

With the rapid development of AI technology, the energy consumption becomes increasingly important. Take Chat GPT as an example, according to publicly available data, it is known that the competing ability of GPT-4 achieved a 20-fold improvement over its predecessor. The power usage of data centers has become a significant concern with the widespread application of AI and the proliferation of data centers. Especially, in the field of deep learning, the training and inference processes of models demand substantial computational resources. In pursuit of higher accuracy and faster speeds, researchers continuously refine algorithms, resulting in increasingly complex models. While these complex models bring performance improvements, however, they also lead to a sharp rise in energy consumption, especially during the training phase, which involves thousands upon thousands of iterative calculations. At the same time, chip manufacturers are constantly launching more powerful processors to meet the demands of AI technology in various scenarios. These processors increase computing power but also rise in power consumption. Although chip manufacturers have made some gains in energy efficiency, the overall energy consumption still shows an upward trend. In this regard, the sustainability of the AIDC is becoming more and more important.

II. What is AI Data Center

AIDC is a special data center for AI computing that has specialized facilities to meet the special demands of AI workloads. Equipped with high-performance computing resources, these AIDC can handle the complex algorithms and large datasets that are very essential to AI applications. They need sustainable networking infrastructure to ensure fast data transfer, high-capacity storage solutions to accommodate massive data volumes, and energy-efficient designs to manage the high power consumption typical of AI processing. This enables organizations to develop, train, and deploy AI models that can drive innovation and efficiency across various industries.

Therefore, one of the important parts in AIDC is computing power. AI workloads often require significant computing power, especially for training machine learning models. In AIDC typically servers with GPUs (Graphics Processing Units) or TPUs (Tensor Processing Units) that are optimized for parallel processing, which is basis for AI tasks. In this regard, the AIDC infrastructure has to support these servers to work reliable and efficient.

Due to AI workloads are very energy-intensive, AIDC often focus on environmental sustainability including energy efficiency to reduce Greenhouse Gas (GHG) emissions. This includes advanced power supply system, cooling systems, monitoring system, renewable energy sources, and energy management system etc.

The characteristics of supercomputing centers, traditional data centers and AIDCs, are detailed in

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Table 1.

TABLE 1. Different characteristics of AIDCs, Supercomputing Centers, traditional Data Centers and AIDC

Type	Supercomputing Centers	Traditional Data Centers	AIDCs
Construction Objective	Provide support services for scientific computing scenarios	Provide cloud services, storage, and computing services to users	Provide cloud services and computing services, and AI industrial platforms
Specific Functions	Large-scale scientific and engineering computing tasks	Satisfy diverse business application needs of users such as enterprises and governments	Provide computing power and cloud services, and AI industrial platforms
Technical Standards	Multi-use parallel architecture, different technical routes, less interconnection Interoperability	Interconnection and interoperability, open construction	Comprehensive planning, interconnection and interoperability, Open construction
Application Areas	Requiring high computing power	Multiple application scenarios, multiple application fields, multi-disciplinary applications, high network requirements	Typical AI scenarios , requiring high computing power and network capabilities
Application Software	MPI parallel applications	OA systems, business systems, business software	AI models, algorithms, data processing

The high power consumption of severs push the urgent demands on AI data center infrastructure. Firstly, the transition from air- and water-cooling to liquid-cooling necessitates adaptations in cooling systems. Secondly, efficiency and miniaturization are crucial in power supply design. Furthermore, the new layout and load-bearing requirements emphasize the need for adaptable and flexible spaces in AIDC.

III. SUSTAINABLE DEPOLYMENT OF POWER SUPPLY SYSTEM IN AI DATA CENTER

Due to the high density of the sever rack, it is necessary to design highly efficient architecture of AIDC power system to provide the bigger capacity and higher reliability.

3.1 High efficient architecture of power supply system in AIDC

Different power supply system architectures are applied in the data centers on their own conditions such as room height, area etc. Including traditional 1+1 AC Uninterruptable Power Supply (UPS) system structure, Direct AC Grid + Higher Voltage Direct Current (HVDC) power supply system structure, Distributed Power Supply (DPS) system Structure and 12V rack power supply system due to different scenarios [1].

Additionally, it was developed the efficient architecture of power supply system which is 10kV input AC UPS system. The structure of the system was shown as in Figure 1 [2].

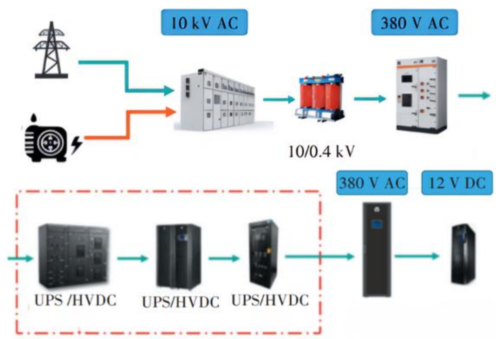

Fig. 1 10kV input AC UPS output system architecture

The current power supply efficiency of AC systems in data centers approximates 95%-96%. Integrating 10KV AC UPS in an optimized mode elevates efficiency by 1%. To optimize further, the ECO mode utilizes bypass for full load supply, swiftly transitioning to battery-backed primary circuit upon bypass failure. Traditional UPS ECO modes risked untimely switching due to cold standby inverters. New designs incorporate hot standby inverters, facilitating timely ECO-to-primary transition within specified limits upon bypass malfunction.

3.2 Sustainable deployment case of power supply system in AIDC

The new design dedicated to the 10KV input AC ups was realized in this AIDC project. Here also presented some new design due to AI racks in other parts of power supply system so as to make sure the operation of Generative AI or Inferential AI racks more sustainable, green, and efficient.

The sustainable design of 10kV input AC UPS： Four 10kV power stations are connected in this AIDC pilot project, with each pair of 10kV power sources mutually serving as primary and backup. Two sets of 10kV high-voltage systems are constructed and installed separately in power supply rooms. The 10kV power distribution system employs a double-bus sectionalized wiring configuration and automatic transfer switch is installed to ensure the reliability and efficiency of the power supply system..

The new design on transformer： Given the substantial power consumption demands and the confined space available for power supply room renovations, a tailored SCB18 3150kVA transformer had been selected and installed. On the primary transformer's low-voltage side, a segmented single busbar design was employed, which ensured continuity of power supply by maintaining at least one operational power path during maintenance or repair activities conducted on any part of the busbar on the same transformer's low-voltage side.

The sustainable design on low voltage distribution system： A power supply system that integrated transformers, low-voltage cabinets, UPS units, and essential components into a unified equipment of 10kV input AC UPS was realized and significantly reduced the space requirements. In addition, by minimizing extensive cabling, this new design decreased routing losses and enhanced the overall energy efficiency of the whole power supply system so as to promote the sustainable deployment.

The new design on AC UPS: The redundant 4+1 and 3+1 parallel UPS were applied by an external maintenance bypass, ensure a UPS battery system backup of at least 7 minutes. Each UPS is discretely furnished with dedicated input/output switchgear cabinets, enabling seamless maintenance during individual UPS abnormal situations.

The sustainable design on power distribution for IT servers: The power supply architecture for IT cabinets integrated different busbars, with each low-voltage distribution cabinet deploying large capacity busbars transiting to intelligent small busbars (two per cabinet column). This dual-path configuration ensures continuous power to cabinets and reaching higher reliability and efficiency as well as decreasing cabling complexity, optimizing space & resource utilization, alleviates customer maintenance and decrease emergency response burdens.

IV. SUSTAINABLE DEPOLYMENT OF COOLING SYSTEM IN AI DATA CENTER

4.1 Architecture of cooling system in AIDC

(1) Energy-Saving Technologies and Strategies for Water-Cooled Systems

The architecture of the water-cooled air conditioning system in data centers, as depicted in Figure 2, primarily comprises cooling towers, pumps, piping, chillers, indoor terminal devices, and other components. A typical system architecture is illustrated in Figure 2 [3].

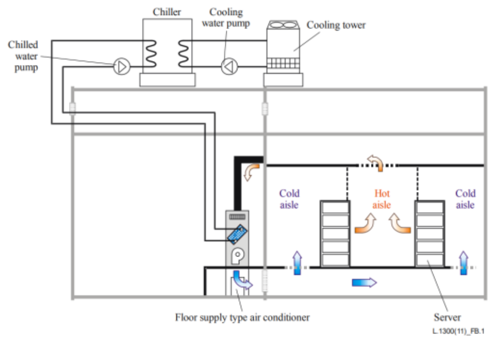

Fig. 2 Air conditioning system architecture in data centers

To address the issues of low cooling efficiency, limited heat dissipation capacity and inability to meet the thermal management requirements of energy- intensive server racks in traditional data centers, several proximal cooling solutions have been raised and realized, such as in-row cooling technology and backdoor cooling technology etc.

The in-row cooling is a data center cooling system where the cooling units are placed directly within the rows of server racks, rather than in a separate room or at the perimeter of the data center. This method of cooling is designed to address the high heat density generated by IT server equipment more efficiently than traditional perimeter or underfloor cooling systems. In-row coolers are situated in close proximity to the heat-generating IT equipment. This allows for a more direct and efficient transfer of heat from the servers to the cooling units, reducing the amount of energy required to cool the data center.

The backdoor cooling for data centers, also known as rear door heat exchangers or rear door cooling, is a cooling method used in data centers to manage the heat generated by server racks. This approach involves installing a heat exchanger or cooling unit directly on the back of the server rack, where the hot air exhaust is released. As servers operate, they generate heat that is exhausted out the back. The rear door cooling unit removes this heat by transferring it from the server exhaust air to a coolant or refrigerant within the unit. By placing the cooling unit at the point of heat generation (the back of the server rack), it can capture and remove heat more efficiently before it enters the general data center environment. This approach enhances energy efficiency, making the system more sustainable and environmentally friendly. By targeting the heat source directly and operating at higher supply air temperatures, it optimizes performance while reducing energy consumption.

（2）Liquid cooling system in AIDC

Recent liquid cooling solutions could be categorized into three different types namely cold plate cooling, immersion cooling (comprising single-phase and phase-change immersion), and spray liquid cooling. Due to Immersion cooling and spray liquid cooling have direct connection to IT equipment, it was classified as contact-based. Conversely, cold plate cooling is non-contact based, as the coolant does not directly interface with heat-generating components. Figure 3 illustrates the whole system structure [4].

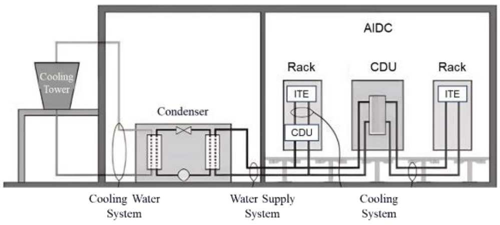

Fig. 3 Schematic diagram of liquid cooling system

Based on the industry experience, it could be summarized from the pilot project cases of various liquid technologies that immersion liquid cooling technology typically requires customized design and modification of IT equipment, with a significant update to the original infrastructure of the data center, so it will make the subsequent maintenance more challenging. Spray-type liquid cooling directly sprays cooling fluid onto heat-generating components. During the spraying process, issues like drift can affect other equipment in the data center. This technology also presents higher technical and maintenance challenges, which currently limits its application. Cold plate liquid cooling technology, on the other hand, is one of the earliest and most widely used liquid cooling methods. It requires relatively less modification of IT equipment and existing data center infrastructure. Its feasibility, technical maturity, safety, reliability, and ease of maintenance are relatively high. In summary, the cold plate liquid cooling solution is a key technology for addressing heat dissipation in energy intensive data centers [5].

4.2 Deployment case of water cooling system in AIDC

In the context of high power density per rack deployment, this AIDC project adopts air-cooled variable frequency fluorine pump air conditioning to address the heat dissipation challenges of huge numbers of server deployment with high density of 35kW per rack. The key points in the specific design are as follows.

The new design on Energy-Efficient Indoor Unit: The precision air conditioners which utilize the most energy-efficient variable frequency fluorine pump technology for air-cooled systems in this project. When the outdoor temperature is low, the system can fully utilize natural cooling sources. The operation of the fluorine pump can replace or partially replace the operation of the compressor from traditional air-conditioning system, so as to reduce the energy consumption of the cooling system. During summer conditions, the compressor is used for continuous cooling, while in transitional seasons and winter, the energy-saving operation of the fluorine pump can be realized.

The new design on Efficient Outdoor Unit Layout: This project utilizes high-efficiency centralized V-type condensers, integrating the fluorine pump within the outdoor unit. It is required that the outdoor units be spaced as far apart as possible to optimize heat dissipation and avoid the heat island effect. More importantly, this arrangement prevents the fluorine pump from being in a high-temperature environment and ensuring that the outdoor units dissipate heat effectively as well as operate in a well-ventilated area so as to maximize the use of natural cooling sources.

The new design of the air conditioning outdoor units and pipelines to reduce attenuation: The space for the air conditioning outdoor units is increased due to the requirement on ensuring effective heat dissipation. Both the design and construction had realized the shortest possible distance between the indoor units and the outdoor units of the air conditioning system, which aims to reduce the attenuation of cooling capacity caused by long pipelines.

The new design on optimized Airflow Management: This project adopts an in-row cooling system with horizontal supply/return airflow and a closed hot aisle solution. Through a well-designed airflow organization, the efficiency of cold source utilization is improved, thereby energy consumption was reduced accordingly. The isolation of hot and cold aisles and airflow barriers are implemented to prevent the mixing of hot and cold air.

The new design on independent temperature and humidity control: The pilot application of wet film humidifiers has installed in this AIDC deployment. It can be concluded that the humidification function alone in the data centers can save 9% of the electricity consumption of the cooling system based on measured data. This eliminates the unnecessary energy consumption caused by the cooling system's own humidification and dehumidification processes which are mutually contradictory.

Finally, these deployment solved the challenges of server rack capacity is as large as 35kW, and it is significantly increasing cooling system efficiency and reaching a target of Power Usage Effectiveness (PUE) could be decreased to less than 1.3.

4.3 Sustainable deployment case of cooling system in AIDC

This deployment case design incorporates cold plate liquid cooling technology with the air conditioning cold source comprising both liquid cooling and water cooling dual systems. The liquid cooling source utilizes an "open cooling tower + plate heat exchanger" approach. The system is divided into cooling side, primary side, and secondary side. The heat from major heat-generating components is dissipated through the cooling fluid circuit in the server's cold plates, with the specific architecture illustrated in Figure 4. Additionally, a centralized CDU arrangement is employed within the data center. An intelligent system has also been established including an environmental monitoring system, a building equipment monitoring system, and a cold source group control system to complete the intelligent management of the AIDC. Importantly, based on different loads, this deployment on cooling system achieves the aim of dynamically adjusting equipment status to realize energy saving and carbon reduction, making the system more sustainable, efficient, and environmentally friendly.

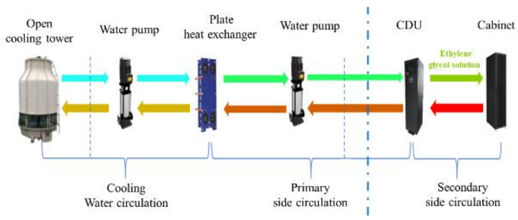

Fig. 4 Example of liquid cooling system deployment

Via using liquid cooling technology, the PUE of this AIDC decreased from 1.312 to 1.233. As an initial design on capacity of this AIDC is 24MW, if it works on full load mode, annal energy saving of this AIDC can reach 16.61 million kWh of electricity.

V. SUSTAINABLE DEPOLYMENT OF MONITORING SYSTEM IN AI DATA CENTER

5.1 Architecture of Data Center Infrastructure Management system in AI data

Data center Infrastructure Management (DCIM) system is a comprehensive suite of software and hardware solutions designed to manage and monitor the physical infrastructure of data centers. DCIM provides tools for the efficient management of data center operations, including power, cooling, physical space, and IT assets. The primary goal of DCIM is to optimize data center performance and efficiency while ensuring reliability and sustainability. It plays a crucial role in creating a more efficient and environmentally friendly data center environment.

DCIM systems normally have the following functionalities:

(1) **Asset Management:** Manage all IT assets, including servers, storage devices, network equipment, and other hardware, along with their configurations, locations, and relationships, as well as track any changes to these assets.

(2) **Capacity Planning:** Monitors and predicts power, cooling, and space capacity to ensure resources are available for current and future needs.

(3) **Energy Management:** Monitors energy consumption and usage efficiency to identify opportunities for energy savings and to control as accordingly.

(4) **Power Management:** Monitors power distribution, usage, and efficiency to prevent failure time and manage power costs.

(5) **Environmental Monitoring:** Monitor environmental conditions such as temperature, humidity, and air flow to maintain optimal conditions for data center equipment.

(6) **Cooling Management:** Ensures that cooling resources are used efficiently to maintain the appropriate temperature for all the equipment in data center.

(7) **Visualization:** Provides visualization tools, including 3D modeling, floor mapping, and rack diagrams to help data center managers understand the layout and status of their

infrastructure.

(8) **Reporting and Analytics:** Generates reports and provides analytics to support decision-making, including energy usage, capacity forecasting, and cost analysis.

It is also possible DCIM provides some other functions which are not mentioned in above based on there different requirements from the load.

The typical structure of DCIM system is shown in Fig.5 [6].

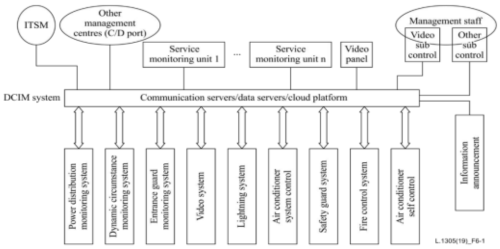

Figure 5 An example on the structure of DCIM system

5.2 Deployment case of DCIM in AIDC

In this deployment case for DCIM in AIDC, AI technology is also applied to save more energy. The system constructs a unified, intelligent, and digitalized energy operation management platform that transcends disciplines, platforms, and diverse supply enterprises, leveraging the Lambda big data architecture along with the AIDC monitoring system and the smart energy brain as in figure 6.

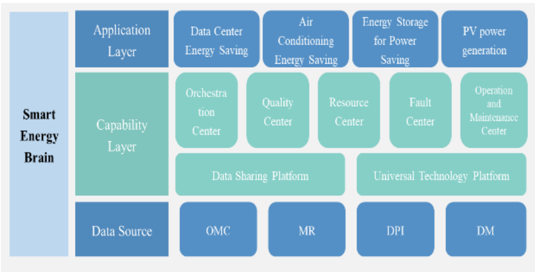

Figure 6 Structure of smart energy brain in AIDC

This DCIM system provides a one-stop comprehensive operational management solution for network energy, relying on big data computing, AI algorithm operations, and micro-service management to build four energy-saving units. It also introduces an orchestration center, a fault center, and a resource center to achieve command distribution and monitoring. The entire architecture follows a layered decoupling approach, enabling energy-saving management and control across the entire network and among different manufacturers.

The system is based on big data analysis, identifying business conditions, seasonal differences, and other characteristics to construct diverse energy-saving scenario models. DCIM spans across meteorological networks, main equipment OMC, and environmental monitoring for end-to-end big data mining. Based on clustering analysis models, it builds temperature control models focusing on the cooling capacity and energy efficiency ratio of cooling equipment groups in terms of temperature fields and thermal loads. Additionally, using LSTM algorithms, it achieves the prediction of objective environmental and business demand parameters in advance. Moreover, based on a proprietary neural network algorithm, it performs intelligent optimization for precise matching of cooling supply and demand, temperature field boundary constraints, and structured power consumption, obtaining the optimal energy-saving strategy for air conditioning and executing it. Through continuous evaluation of cooling quality and energy-saving indicators, along with intelligent iteration of algorithms, the system further enhances the accuracy of the models and expands their application scope, achieving comprehensive and intelligent energy-saving for air conditioning.

VI. ANALYSIS ON INTERNATIONAL STANDARDS RELATING TO AI DATA CENTER

The international standards on technical specifications and energy efficiency metrics requirements of whole data center as well as its sub-systems are analyzed in the following content. Especially, these metrics are very essential tools for AI data center to evaluate their energy efficiency and implement improvements accordingly.

6.1 Analysis on International standards on power supply system in Data Center

The basic requirement to Up to 400V DC system for data center is shown in ITU-T L.1200 “Direct current power feeding interface up to 400 V at the input to telecommunication and ICT equipment” and IEC 62040-5-3-2016 Uninterruptible power systems (UPS)–Part 5-3: DC output UPS- performance and testing requirements. The structure of typical power supply system structure “Direct AC grid + HVDC” system was defined in ITU-T L.1204 as Figure 7[7].

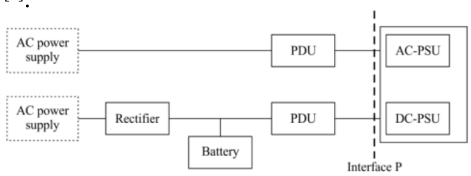

Figure 7 Direct AC grid + HVDC power supply system

The power system energy efficiency could be measured according to the ITU-T L.1320, it includes AC UPS, HVDC system, inverters, rectifiers and DC/DC converters energy efficiency metrices and methodology[8]. Additionally, for the data center which supplied from renewable energy supply, it's Renewable Energy Factor (REF) that indicates the proportion of the renewable energy in the whole data center energy supply should be calculated as follows[9].

$R_{EF}= E_{ren}/ E_{total}$………… ……………..(1)

Where:

E_{ren} is energy supply capacity from renewable energy system.

E_{total} is the total data centre energy consumption (annual), in kWh;

The energy efficiency of entire power supply system of data center $\eta_{Whole\_DC\_POWER}$, is calculated with following formula [10].

$PLF = E_{power\text{-}consumption}/E_{IT} = (E_{input\_power} - E_{IT})/ E_{IT}$………… ………… …………..(2)

Where:

E_{IT} is the IT equipment energy consumption (annual), in kWh;

$E_{power\text{-}consumption}$ is the energy consumption from entire power supply system in data center;

6.2 Analysis on International standards on cooling system in Data Center

The cooling system energy efficiency could be measured according to the ITU-T L.1320, and it includes traditional air conditor, outdoor air system and heat exchanging system energy efficiency metrices and methodologies[8]. In addition, ISO/IEC 30134-7 provides the Cooling Efficiency Ratio (CER) and cooling efficiency ratio (R_{CE}) of cooling system which represent the ratio of total heat removed and electrical energy used by a cooling system with the formula (3)[11].

$R_{CE}=E_{heat}/E_{cooling}$………… ……………..(3)

Where:

$E_{heat}=E_{heat,DC;}$

$E_{cooling}= E_{cooling,DC}$

$E_{DC}=E_{IT}+ E_{losses} + E_{cooling,DC}$

If available, E_{losses} shall include all other electrical losses, e.g. electrical energy of UPS, energy storage, transformers, power cables or lighting transferred to heat within the DC boundaries.

$E_{cooling,DC}$ is the part of the energy use for the entire shared cooling system to remove the DC related heat loads;

E_{DC} is the total data center energy consumption (annual), in kWh;

E_{IT} is the same with previous definition in formular (2).

6.3 Analysis on International standards on DICM system in Data Center

Based on the management interface of the DCIM system, the five aspects that need to be monitored by a DCIM system are as follows [6].

(1) Power supply devices: it includes high-voltage power distribution parts, low-voltage power distribution parts, voltage transformers, stand-by power supply systems, power transformers and other kinds of electric devices for an IT cabinet. The DCIM system is used to monitor operating conditions and power consumption of the system.

(2) Cooling devices: it includes various unit-type of cooling devices, such as cooling towers of central air conditioners, cool-water machine sets and terminal machine sets of indoor air conditioners.

(3) Data center environment: it includes parameters such as the temperature, relative humidity, air flow and hot spots (if present) are monitored by the DCIM system as environmental aspects. Moreover, the monitoring area not only covers the room, but also equipment cabinets and the module data center area. It also includes monitoring video and surveillance systems.

(4) Lightning system: there are different types of lightning protections based on the management requirements of the data center. As a prerequisite of lightning for management, the working mode should be adjusted automatically to reduce the power consumption and working load.

(5) Fire control system: information on the fire control system and status of the fire control system should be transferred to the DCIM system so that there will be in-time protection through the system.

In addition, requirements on maintenance work, intelligence strategy on prediction and working principles for data center energy efficiency optimization were raised in this Recommendation.

6.4 Analysis on International standards on whole Data Center

One of the commonly used parameters for measuring energy efficiency is PUE, which is defined in ISO/IEC 30134-2. The calculation of the PUE is shown as in formula (4). It aims to reflect if supporting facilities efficiency [12].

$PUE = E_{DC} /E_{IT}$……… ……………..(4)

where

E_{DC} is the same with previous definition in formular (3);

E_{IT} is the same with previous definition in formular (2).

Additionally, ITU-T L.1302 also provides assessment methodology includes an assessment methodology of the energy efficiency for the whole data centre/telecom centre infrastructure (or facility), for a partial data centre/telecom centre infrastructure (or facility) [11].

Water usage is another important point in data center. The parameter named Water Usage effectiveness (WUE) was defined as following formula in IEC/ISO 30134-9 standard, it is ratio of the data centre water consumption divided by the energy consumed by IT equipment as shown in formular (5)[13].

$\eta_{U,W}= Uw/E_{IT}$ ………. …………………..(5)

Annual water usage is calculated as following formula.

$Uw=Iw-Ow$……. ………. ……. …………..(6)

Where:

$\eta_{U,W}$: Water utilization efficiency.

Uw Annual water usage of data center (m^3).

Iw : total water input from outside the data centre boundaries (annual) measured by total volume in m^3.

Ow : total water returned out of the data centre boundaries (annual) measured by total volume in m^3.

E_{IT} is the same with previous definition in formular (2);

Additionally, from the point of carbon emissions, the data center could have Carbon Usage Effectiveness as defined in IEC/ISO 30134-7, it was shown as formula (7) [14].

$\eta_{U,C}= C_{DC}/ E_{IT}$ ……. ……. ……………..(7)

Where:

$\eta_{U,C}$: Carbon usage effectiveness.

C_{DC} is CO_2 emissions of the data centre in kg;

E_{IT} is the same with previous definition in formular (2).

VII. Conclusion

This paper addresses the energy consumption challenges of AIDC and analyzes its various subsystems. Firstly, it focuses on the emerging power supply technologies, including the 10kV input AC UPS architecture and provides case studies on their application in AI data center design. Secondly, it discusses recent technology in data center cooling system, particularly liquid cooling, and explores its deployment in AIDC. Thirdly, the paper presents the architecture of Data Center Infrastructure Management (DCIM) systems and a practical deployment case. Finally, this paper highlights international standardization work related to technical specifications and energy efficiency, covering both the aforementioned areas and the entire AIDC.

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